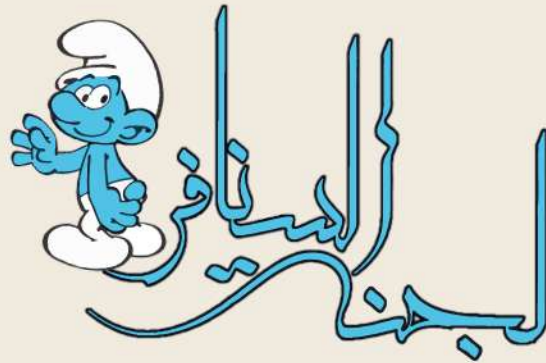
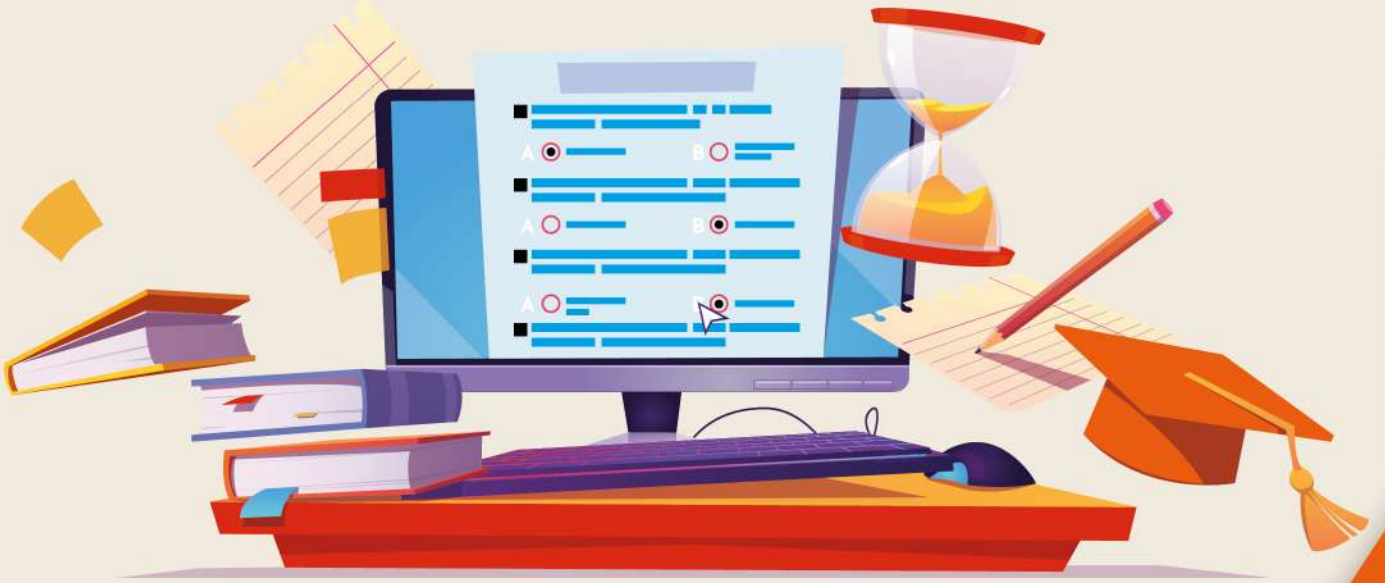


خدمتكم عبادة نتقرب بها الى الله

2021

أسئلة سنوات ورقي ميد الفيزياء العامة 2



سنافر البوليتكنك 2021





@Useful Information@

Permittivity of free space = $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$, Mass of the Proton = $M_p = 1.67 \times 10^{-27} \text{ Kg}$.
Coulomb's Constant = $k_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$, Charge of the electron = $-1.602 \times 10^{-19} \text{ C}$.
Mass of the electron = $M_e = 9.11 \times 10^{-31} \text{ Kg}$, Acceleration due to gravity = $g = 9.8 \text{ m/s}^2$

Fill your answer in the following table:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	A	D	C	A	B	A	C	A	B
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
D	E	E	D	B	C	B	D	C	B

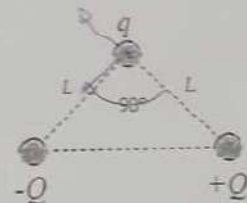
Choose the correct answer:

- Q1. Charge of uniform density (40 pC/m^2) is distributed on a spherical surface (radius = 1.0 cm), and a second concentric spherical surface (radius = 3.0 cm) carries a uniform charge density of 60 pC/m^2 . What is the magnitude of the electric field at a point 5.0 cm from the center of the two surfaces?
a) 3.8 N/C b) 4.1 N/C c) 3.5 N/C d) 2.6 N/C
- Q2. If the electric field just outside a thin non-conducting sheet is equal to 2.5 N/C , determine the surface charge density on the sheet.
a) 27 pC/m^2 b) 53 pC/m^2 c) 44 pC/m^2 d) 13 pC/m^2
- Q3. A point charge of 4.0 nC is placed at the center of a hollow spherical conductor (inner radius = 1.0 cm , outer radius = 2.0 cm) which has a net charge of -4.0 nC . Determine the resulting charge density on the inner surface of the conducting sphere.
a) $+4.8 \text{ } \mu\text{C/m}^2$ b) $-3.2 \text{ } \mu\text{C/m}^2$ c) $-9.5 \text{ } \mu\text{C/m}^2$ d) $+9.5 \text{ } \mu\text{C/m}^2$
- Q4. A particle (charge = $30 \text{ } \mu\text{C}$) moves in a region where the only force on it is an electric force. As the particle moves 25 cm from point A to point B, its kinetic energy increases by 1.5 mJ . Determine the magnitude of electric potential difference, $|V_B - V_A|$.
a) $+50 \text{ V}$ b) $+40 \text{ V}$ c) $+30 \text{ V}$ d) $+60 \text{ V}$
- Q5. Suppose the radius of an insulated sphere is decrease .What happens to the flux and the magnitude of the electric field at the surface of the sphere?
a) The flux and the field both increase.
b) The flux increases and field decreases.
c) The flux remains the same and the field increases.
d) The flux decreases and field remains the same.

- # Q6. A uniformly charged ring of radius 15 cm has a total charge of $20 \mu\text{C}$. The electric field on the axis of the ring at 15 cm.
 a) $5.42 \times 10^6 \text{ N/C}$ b) $4.24 \times 10^6 \text{ N/C}$ c) $2.83 \times 10^6 \text{ N/C}$ d) $3.24 \times 10^6 \text{ N/C}$
- Q7. A closed surface contains the following point charges: $-9 \mu\text{C}$, $-5 \mu\text{C}$, $3 \mu\text{C}$, $4 \mu\text{C}$. The electric flux through the surface of the box (in units $\text{N} \cdot \text{m}^2/\text{C}$) is?
 a) 4.52×10^5 b) -7.91×10^5 c) 4.52×10^5 d) 7.91×10^5
- Q8. Charge of uniform density (40 pC/m^2) is distributed on a spherical surface (radius = 1.0 cm), and a second concentric spherical surface (radius = 3.0 cm) carries a uniform charge density of 60 pC/m^2 . What is the magnitude of the electric field at a point 5.0 cm from the center of the two surfaces?
 a) 3.8 N/C b) 4.1 N/C c) 3.5 N/C d) 2.6 N/C
- Q9. If the electric field just outside a thin non-conducting sheet is equal to 2.5 N/C , determine the surface charge density on the sheet.
 a) 27 pC/m^2 b) 53 pC/m^2 c) 44 pC/m^2 d) 13 pC/m^2
- Q10. A point charge of 4.0 nC is placed at the center of a hollow spherical conductor (inner radius = 1.0 cm, outer radius = 2.0 cm) which has a net charge of -4.0 nC . Determine the resulting charge density on the inner surface of the conducting sphere.
 a) $+4.8 \mu\text{C/m}^2$ b) $-3.2 \mu\text{C/m}^2$ c) $-9.5 \mu\text{C/m}^2$ d) $+9.5 \mu\text{C/m}^2$
- Q11. A particle (charge = $30 \mu\text{C}$) moves in a region where the only force on it is an electric force. As the particle moves 25 cm from point A to point B, its kinetic energy increases by 1.5 mJ. Determine the magnitude of electric potential difference, $|V_B - V_A|$.
 a) +50 V b) +40 V c) +30 V d) +60 V
- Q12. Suppose the radius of an insulated sphere is decrease. What happens to the flux and the magnitude of the electric field at the surface of the sphere?
 a) The flux and the field both increase.
 b) The flux increases and field decreases.
 c) The flux remains the same and the field increases.
 d) The flux decreases and field remains the same.
- Q13. The electric potential at $x = 3 \text{ m}$ is 130 V, and the electric potential at $x = 5 \text{ m}$ is 220 V, assuming the field is uniform. The magnitude of the x-component of the electric field in this region is:
 a) 50 N/C b) 45 N/C c) 35 N/C d) 30 N/C
- Q14. A solid conducting sphere (radius = 5.0 cm) has a charge of 0.20 nC distributed uniform surface. If point A is located at the center of the sphere and point B is 15 cm from the center, what is the magnitude of the electric potential difference between these two points?
 a) 24 V b) 30 V c) 15 V d) 45 V
- Q15. When a negative charge is released and moves along an electric field line, it moves position of:
 a) lower potential and lower potential energy.
 b) lower potential and higher potential energy.
 c) higher potential and lower potential energy.
 d) higher potential and higher potential energy.

- Q15. An object with negative charge is placed in a region of space where the electric field is directed vertically upward. What is the direction of the electric force exerted on this charge?
 a) It is up. b) It is down. c) There is no force. d) The force can be in any direction.

- Q16. If $Q = 25 \mu\text{C}$, $q = 10 \mu\text{C}$, and $L = 30 \text{ cm}$ in the figure, what is the magnitude of the electrostatic force on q ?
 a) 35 N b) 22 N c) 20 N d) 14 N



- Q17. A linear charge of uniform density equal to 8.0 nC/m is distributed along the x axis from $x = -3.0 \text{ m}$ to $x = 3.0 \text{ m}$. What is the magnitude of the electric field at the point $x = 6.0 \text{ m}$ on the x axis?
 a) 60 N/C b) 16 N/C c) 26 N/C d) 15 N/C

- Q18. A particle (mass = 6.0 g , charge = 40 mC) moves in a region of space where the electric field is uniform and is given by $E_x = -5.5 \text{ N/C}$, $E_y = E_z = 0$. If the position and velocity of the particle at $t = 0$ are given by $x = y = 0$ and $v_x = 50 \text{ m/s}$ along +ve x -axis, $v_y = v_z = 0$, what is the distance from the origin to the particle at $t = 2.0 \text{ s}$?
 a) 60 m b) 28 m c) 44 m d) 27 m

- Q19. A uniformly charged ring of radius 15 cm has a total charge of $20 \mu\text{C}$. The electric field on the axis of the ring at 15 cm .
 a) $5.42 \times 10^6 \text{ N/C}$ b) $4.24 \times 10^6 \text{ N/C}$ c) $2.83 \times 10^6 \text{ N/C}$ d) $3.24 \times 10^6 \text{ N/C}$

- Q20. A closed surface contains the following point charges: $-9 \mu\text{C}$, $-5 \mu\text{C}$, $3 \mu\text{C}$, $4 \mu\text{C}$. The electric flux through the surface of the box (in units $\text{N} \cdot \text{m}^2/\text{C}$) is?
 a) 4.52×10^5 b) -7.91×10^5 c) 4.52×10^5 d) 7.91×10^5

Good Luck



#Useful Information#

Permittivity of free space = $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$.
Coulomb's Constant = $k_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$.
Mass of the electron = $M_e = 9.11 \times 10^{-31} \text{ Kg}$.

Mass of the Proton = $M_p = 1.67 \times 10^{-27} \text{ Kg}$.
Charge of the electron = $-1.602 \times 10^{-19} \text{ C}$.
Acceleration due to gravity = $g = 9.8 \text{ m/s}^2$

Fill your answer in the following table:

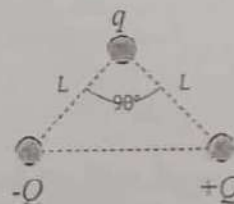
Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
d	b	a	c	a	b	b	c	c	b
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
b	d	b	d	b	d	d	d	b	d

Choose the correct answer:

Q1. A point charge Q is placed on the x axis at $x = 2.0 \text{ m}$. A second point charge, $-Q$ is placed at $x = 4.0 \text{ m}$. If $Q = 40 \mu\text{C}$, what is the magnitude of the electrostatic force on a $30 \mu\text{C}$ charge placed at the origin?
a) 7.2 N b) 3.9 N c) 1.5 N d) 2.0 N

Q2. An object with negative charge is placed in a region of space where the electric field is directed vertically upward. What is the direction of the electric force exerted on this charge?
a) It is up. b) It is down. c) There is no force. d) The force can be in any direction.

Q3. If $Q = 25 \mu\text{C}$, $q = 10 \mu\text{C}$, and $L = 30 \text{ cm}$ in the figure, what is the magnitude of the electrostatic force on q ?
a) 35 N b) 22 N c) 20 N d) 14 N



Q4. A linear charge of uniform density equal to 8.0 nC/m is distributed along the x axis from $x = -3.0 \text{ m}$ to $x = 3.0 \text{ m}$. What is the magnitude of the electric field at the point $x = 6.0 \text{ m}$ on the x axis?
a) 60 N/C b) 16 N/C c) 26 N/C d) 15 N/C

Q5. A particle (mass = 6.0 g , charge = 40 mC) moves in a region of space where the electric field is uniform and is given by $E_x = -5.5 \text{ N/C}$, $E_y = E_z = 0$. If the position and velocity of the particle at $t = 0$ are given by $x = y = 0$ and $v_x = 50 \text{ m/s}$ along +ve x -axis, $v_y = v_z = 0$, what is the distance from the origin to the particle at $t = 2.0 \text{ s}$?
a) 60 m b) 28 m c) 44 m d) 27 m

Q6. The electric potential at $x = 3$ m is 130 V, and the electric potential at $x = 5$ m is 220 V, assuming the field is uniform. The magnitude of the x-component of the electric field in this region is:
 a) 50 N/C b) 45 N/C c) 35 N/C d) 30 N/C

Q7. A solid conducting sphere (radius = 5.0 cm) has a charge of 0.20 nC distributed uniform surface. If point A is located at the center of the sphere and point B is 15 cm from the center, what is the magnitude of the electric potential difference between these two points?
 a) 24 V b) 30 V c) 15 V d) 45 V

Q8. When a negative charge is released and moves along an electric field line, it moves position of:
 a) lower potential and lower potential energy.
 b) lower potential and higher potential energy.
 c) higher potential and lower potential energy.
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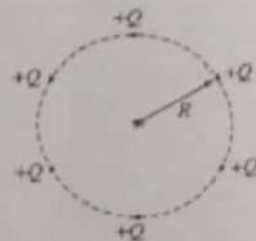
Q9. A rod (length = 2.0 m) is uniformly charged and has a total charge of 6.0 nC. What is electric potential (relative to zero at infinity) at a point which lies along the axis of the rod and is 3.0 m from the center of the rod?
 a) 22 V b) 19 V c) 16 V d) 25 V

Q10. The electric flux (in $\text{N}\cdot\text{m}^2/\text{C}$) of the electric field $24\mathbf{i} + 30\mathbf{j} + 16\mathbf{k}$ N/C through a 2 m^2 portion of the xy-plane is:
 a) 42 b) 48 c) 34 d) 32

Q11. Object A has a charge of $+3\mu\text{C}$, and object B has a charge of $+9\mu\text{C}$. Which statement is true about the electric forces on the objects?
 a) $F_{AB} = -3F_{BA}$ b) $F_{AB} = -F_{BA}$ c) $3F_{AB} = -F_{BA}$ d) $F_{AB} = 3F_{BA}$

Q12. A non-conducting sphere of radius 3 cm has a charge of $6\mu\text{C}$ distributed uniformly throughout its volume. What is the magnitude of the electric field within the sphere 1.5 cm from its center (in units N/C)?
 a) 1×10^7 b) 6×10^7 c) 5×10^7 d) 3×10^7

Q13. A six equal point charges (each with charge $Q = 3\text{ nC}$) are placed symmetrically around a circle of radius $R = 20$ cm. How much work would be required to bring a seventh particle of charge $+Q$ from very far away and place it at the center of the circle?
 a) $4.86\mu\text{J}$ b) $6.48\mu\text{J}$ c) $3.24\mu\text{J}$ d) $2.43\mu\text{J}$



Q14. A point charge Q is placed on the x axis at $x = 2.0$ m. A second point charge, $-Q$ is placed at $x = 4.0$ m. If $Q = 40\mu\text{C}$, what is the magnitude of the electrostatic force on a $30\mu\text{C}$ charge placed at the origin?
 a) 7.2 N b) 3.9 N c) 1.5 N d) 2.0 N

- Q16. A rod (length = 2.0 m) is uniformly charged and has a total charge of 5.0 nC. What is electric potential (relative to zero at infinity) at a point which lies along the axis of the rod and is 3.0 m from the center of the rod?

a) 22 V b) 19 V c) 16 V d) 25 V

- Q17. The electric flux (in $\text{N}\cdot\text{m}^2/\text{C}$) of the electric field $24\mathbf{i}+30\mathbf{j}+16\mathbf{k}$ N/C through a 2 m^2 portion of the yz-plane is :

a) 42 b) 32 c) 34 d) 48

- Q18. Object A has a charge of $+2\mu\text{C}$, and object B has a charge of $+6\mu\text{C}$. Which statement is true about the electric forces on the objects?

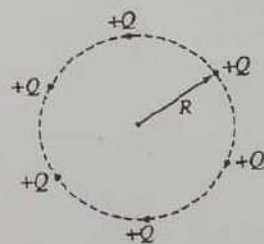
a) $F_{AB} = -3F_{BA}$ b) $F_{AB} = -F_{BA}$ c) $3F_{AB} = -F_{BA}$ d) $F_{AB} = 3F_{BA}$

- Q19. A non-conducting sphere of radius 3 cm has a charge of $2\mu\text{C}$ distributed uniformly throughout its volume. What is the magnitude of the electric field within the sphere 1.5 cm from its center (in units N/C)?

a) 1×10^7 b) 6×10^7 c) 5×10^7 d) 3×10^7

- Q20. A six equal point charges (each with charge $Q=3\text{ nC}$) are placed symmetrically around a circle of radius $R=10\text{ cm}$. How much work would be required to bring a seventh particle of charge $+Q$ from very far away and place it at the center of the circle?

a) $4.86\mu\text{J}$ b) $6.48\mu\text{J}$ c) $3.24\mu\text{J}$ d) $2.42\mu\text{J}$



Good Luck

Useful Information

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Mass of the electron = $M_e = 9.11 \times 10^{-31} \text{ Kg}$.

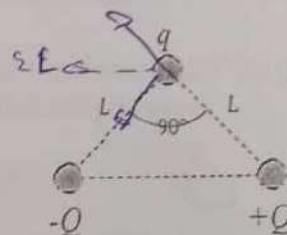
Mass of the Proton = $M_p = 1.67 \times 10^{-27} \text{ Kg}$.
Charge of the electron = $-1.602 \times 10^{-19} \text{ C}$.
Acceleration due to gravity = $g = 9.8 \text{ m/s}^2$

Fill your answer in the following table:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	A	B	C	C	B	D	B	A	B
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
A	C	A	B	B	A	D	B	C	A

Choose the correct answer:

- Q1. A point charge Q is placed on the x axis at $x = 2.0 \text{ m}$. A second point charge, $-Q$ is placed at $x = 3.0 \text{ m}$. If $Q = 40 \mu\text{C}$, what is the magnitude of the electrostatic force on a $30 \mu\text{C}$ charge placed at the origin?
a) 7.2 N b) 3.9 N **c) 1.5 N** d) 14 N
- Q2. An object with positive charge is placed in a region of space where the electric field is directed vertically upward. What is the direction of the electric force exerted on this charge?
a) It is up. b) It is down. c) There is no force. d) The force can be in any direction.
- Q3. If $Q = 25 \mu\text{C}$, $q = 10 \mu\text{C}$, and $L = 40 \text{ cm}$ in the figure, what is the magnitude of the electrostatic force on q ?
a) 28 N **b) 22 N** c) 20 N d) 14 N
- Q4. A linear charge of uniform density equal to 8.0 nC/m is distributed along the x axis from $x = -2.0 \text{ m}$ to $x = 3.0 \text{ m}$. What is the magnitude of the electric field at the point $x = 6.0 \text{ m}$ on the x axis?
a) 60 N/C b) 10 N/C **c) 26 N/C** d) 15 N/C
- Q5. A particle (mass = 5.0 g , charge = 40 mC) moves in a region of space where the electric field is uniform and is given by $E_x = -5.5 \text{ N/C}$, $E_y = E_z = 0$. If the position and velocity of the particle at $t = 0$ are given by $x = y = 0$ and $v_x = 50 \text{ m/s}$ along +ve x -axis, $v_y = v_z = 0$, what is the distance from the origin to the particle at $t = 2.0 \text{ s}$?
a) 60 m b) 28 m **c) 44 m** d) 12 m



Q16. A rod (length = 2.0 m) is uniformly charged and has a total charge of 6.0 nC. What is electric potential (relative to zero at infinity) at a point which lies along the axis of the rod and is 3.0 m from the center of the rod?
 a) 22 V b) 19 V c) 16 V d) 25 V

Q17. The electric flux (in $\text{N}\cdot\text{m}^2/\text{C}$) of the electric field $24\mathbf{i}+30\mathbf{j}+16\mathbf{k}$ N/C through a 2 m^2 portion of the xy-plane is:

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Q18. Object A has a charge of $+3\mu\text{C}$, and object B has a charge of $+9\mu\text{C}$. Which statement is true about the electric forces on the objects?

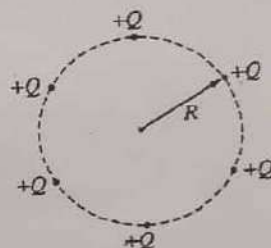
- a) $\mathbf{F}_{AB} = -3\mathbf{F}_{BA}$ b) $\mathbf{F}_{AB} = -\mathbf{F}_{BA}$ c) $3\mathbf{F}_{AB} = -\mathbf{F}_{BA}$ d) $\mathbf{F}_{AB} = 3\mathbf{F}_{BA}$

Q19. A non-conducting sphere of radius 3 cm has a charge of $6\mu\text{C}$ distributed uniformly throughout its volume. What is the magnitude of the electric field within the sphere 1.5 cm from its center (in units N/C)?

- a) 1×10^7 b) 6×10^7 c) 5×10^7 d) 3×10^7

Q20. A six equal point charges (each with charge $Q=3\text{ nC}$) are placed symmetrically around a circle of radius $R=20\text{ cm}$. How much work would be required to bring a seventh particle of charge $+Q$ from very far away and place it at the center of the circle?

- a) $4.86\mu\text{J}$ b) $6.48\mu\text{J}$ c) $3.24\mu\text{J}$ d) $2.43\mu\text{J}$



Good Luck

D Note : let $g=10 \text{ m/s}^2$, $\epsilon_0=8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$, $k_e=9 \times 10^9 \text{ N.m}^2/\text{C}^2$

-Circle the best answer of the following:-

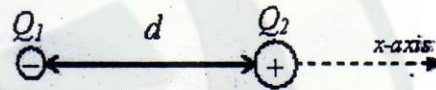
1. (2 points) Point charge of $Q=16 \text{ } \mu\text{C}$ located at a point ($d=20\text{cm}$) along the long axis of a rod of length 20 cm , the rod has a uniform linear charge density $\lambda=36 \text{ } \mu\text{C/m}$, then the electric force between the rod and the point charge is:



- a) $1.62 \times 10^7 \text{ N}$ b) $8.1 \times 10^5 \text{ N}$ c) 6.5 N d) 13.0 N

2. (2 points)

Two point charges lie along the x-axis as shown in figure, $Q_1 = -4 \text{ nC}$ is located at $x=0 \text{ m}$, and $Q_2 = 16 \text{ nC}$ located at $x=1 \text{ m}$, then the point at which the electric field is zero is located at



- a) $x=0.4 \text{ m}$ b) $x=-1 \text{ m}$ c) $x=1.4 \text{ m}$ d) $x=0.6 \text{ m}$ e) $x=-2 \text{ m}$

3. (2 points) A large, flat, horizontal sheet of charge has a charge per unit area of 0.2 nC/m^2 . Find the electric field just above the middle of the sheet.

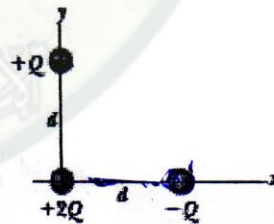
- a) 22.6 N/C b) 28 N/C c) 11.3 N/C d) 17.1 N/C e) 40 N/C

4. (2 points) Charge is uniformly distributed on a very long straight wire. At a distance of 5.0 cm from the wire, the electric field is 800 N/C . What is the charge on a length of 30 cm of the wire?

- a) 1.7 nC b) 0.67 nC c) 2.4 nC d) 1.3 nC e) 0.27 nC

5. (2 points)

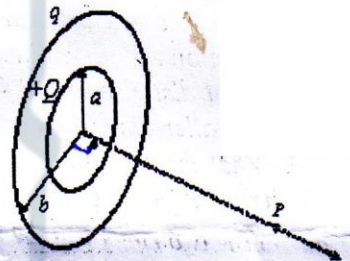
A point charge ($+2Q$) is at the origin and a point charge ($+Q$) is located along the y axis at $y=d$ as in Figure shown. The symbolic expression for the net force vector on a third point charge ($-Q$) located along the x axis at $x=d$.



- a) $\frac{kQ^2}{d^2} \left(\left(-2 + \frac{1}{2\sqrt{2}}\right)\hat{i} + \frac{1}{2\sqrt{2}}\hat{j} \right)$ c) $\frac{kQ^2}{d^2} \left(\left(2 - \frac{1}{2\sqrt{2}}\right)\hat{i} - \frac{1}{2\sqrt{2}}\hat{j} \right)$
b) $\frac{kQ^2}{d^2} \left(\left(2 - \frac{1}{2\sqrt{2}}\right)\hat{i} + \frac{1}{2\sqrt{2}}\hat{j} \right)$ d) $\frac{kQ^2}{d^2} \left(\left(-2 - \frac{1}{2\sqrt{2}}\right)\hat{i} + \frac{1}{2\sqrt{2}}\hat{j} \right)$

6. (2 points)

The figure shows two concentric rings, of radii $a=2R$ and $b=4R$, that lie on the same plane. Point P lies on the central axis, at distance $D=2R$ from the center of the rings. The smaller ring has uniformly distributed charge $+Q$. What must be the uniformly distributed charge (q) on the larger ring if the net electric field at point P due to the two rings is to be zero



- a) $-1.9Q$ b) $-2.02Q$ c) $-3.95Q$ d) $-4.2Q$ e) $-8Q$

7. (2 points) Two identical conducting spheres each having a radius of 0.2 cm are connected by a light, 3 m long conducting wire. A charge of $80 \mu C$ is placed on one of the conductors. Assume the surface distribution of charge on each sphere is uniform. The tension in the wire is

- a) 0 b) 0.4 N c) 3.24 N d) 4.8 N e) 1.6 N

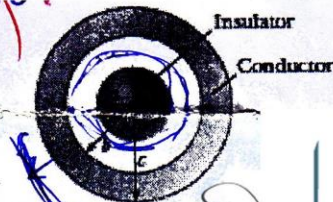
8. (2 points) The electric field at the outer surface of a conducting sphere ($r=20cm$) is equal to $20 \times 10^6 N/m$ then the surface charge density of the sphere is equal to:

final answer $\rightarrow$

$\sigma =$

9. (4 points)

A solid insulating sphere of radius $a=5cm$ carries a net positive charge (16 nC) uniformly distributed throughout its volume. A conducting spherical shell of inner radius $b=10cm$ and outer radius $c=15cm$ is concentric with the solid sphere and carries a net charge ($-32 nC$), fill in the following table the flux Φ through a gaussian sphere of radius r with the same center, and the magnitude of the electric field at r .

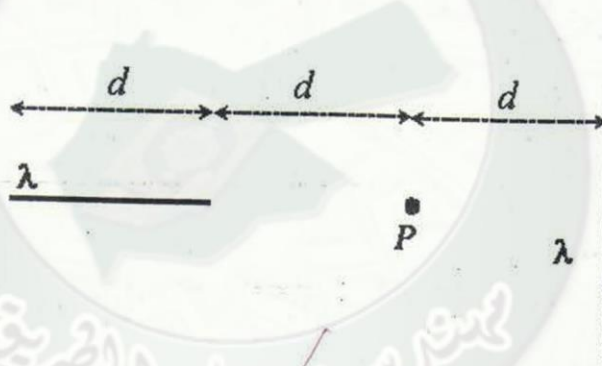


r (cm)	Φ (N.m ² /C)	E (N/C)
2	5785.3	2209 N/C
8	22619.4	220500
12	0	0
16	18,8	585

Note : $k = 9 \times 10^9$, $\epsilon_0 = 8.85 \times 10^{-12}$, electron charge (e^-) = 1.6×10^{-19} C, electron mass (m_e) = 9.11×10^{-31} kg

-Circle the best answer of the following:--

1. In the figure shown two rods with uniform linear charge density (λ), the vertical rod is infinite and the horizontal rod is finite with a length (d), the magnitude of the electric field resulting from both rods at the point (P) is :

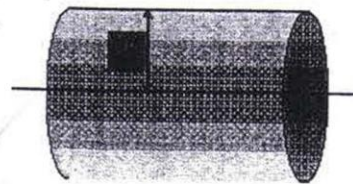


- a) $k\lambda/2d$ b) $5k\lambda/2d$ c) $3k\lambda/2d$ d) $2k\lambda/d$

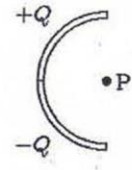
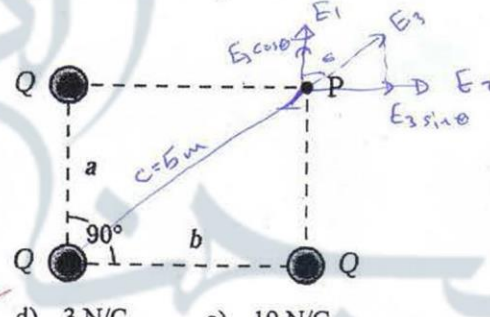
2. A point charge of 8 nC is placed at the center of a hollow spherical conductor (inner radius = 1 cm, outer radius = 2 cm) which has a net charge of -4 nC. At electrostatic equilibrium the charge density (in $\mu\text{C}/\text{m}^2$) of the inner surface of the conducting sphere equal to:

- a) +0.8 b) +6.4 c) -6.4 d) -2.4 e) +2.4

3. A long nonconducting cylinder of radius (R) has a positive charge of a uniform density (ρ) distributed throughout its volume. The magnitude of the electric field at distance r from the cylinder axis, where $r < R$ is given by:-



- a) $\frac{\rho r}{3\epsilon_0}$ b) $\frac{\rho R^3}{2\epsilon_0 r^2}$ c) $\frac{\rho r}{2\epsilon_0}$ d) $\frac{\rho R^2}{2\epsilon_0 r}$ e) $\frac{\rho r^2}{2\epsilon_0}$

4. Positive charge Q is distributed uniformly throughout an insulating sphere of radius R , centered at the origin. A particle with positive charge q is placed at $x = 2R$ on the x axis. The magnitude of the electric field at $x = R/2$ on the x axis is:
- a) $\frac{Q}{4\pi\epsilon_0 R^2}$ b) $\frac{Q}{72\pi\epsilon_0 R^2}$ c) $\frac{Q}{8\pi\epsilon_0 R^2}$ d) $\frac{17Q}{72\pi\epsilon_0 R^2}$
5. A charged oil drop with a mass of 2×10^{-4} kg is held suspended by a downward electric field of 300 N/C. The charge [in μC] on the drop is:
- a) 0 b) -1.5 c) -6.5 d) +1.5 e) +6.5
6. Positive charge $+Q$ is uniformly distributed on the upper half a semicircular rod and negative charge $-Q$ is uniformly distributed on the lower half. What is the direction of the electric field at point P, the center of the semicircle?
- 
- a) $\rightarrow$ b) $\downarrow$ c) $\nearrow$ d) $\uparrow$ e) $\leftarrow$
7. A charge of $+80 \mu\text{C}$ is placed on the x -axis at $x = 0$. A second charge of $-50 \mu\text{C}$ is placed on the x -axis at $x = 50$ cm. the magnitude of the electrostatic force on a third charge of $4 \mu\text{C}$ placed on the x -axis at $x = 30$ cm is equal to:
- a) 77 N b) 25 N c) 39 N d) 13 N e) 45 N
8. If $Q = 16 \text{ nC}$, $a = 3 \text{ m}$, and $b = 4 \text{ m}$, then the magnitude of the electric field at point P shown
- 
- a) 33 N/C b) 13 N/C c) 24 N/C d) 3 N/C e) 19 N/C
9. An electron enters a region of uniform electric field ($E = 50 \text{ N/C}$) with an initial velocity of 40 km/s directed the same as the electric field. After 1.5 ns the speed of the electron is equal to:
- a) 18 km/s b) 42 km/s c) 27 km/s d) 62 km/s e) 53 km/s
10. A uniform linear charge density of 4.0 nC/m is distributed along the entire x axis. Consider a spherical (radius = 5.0 cm) surface centered on the origin. The electric flux (in $\text{N m}^2/\text{C}$) through this surface is equal to:
- a) 68 b) 45 c) 79 d) 62 e) 23



A Name:...

lecture time: 12:15 - 1:15 Instructor : Dr. Ziad Elimat

$$k = \frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N.m}^2/\text{C}^2$$

(Coulomb constant)

$$V_{\text{Sphere}} = \frac{4}{3}\pi r^3$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$$

(Permittivity of free space)

$$V_{\text{Cylinder}} = \pi r^2 h$$

$$g = 9.8 \text{ m/s}^2$$

(Acceleration due to gravity)

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

(Electron mass)

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

(Proton mass)

Prefixes of units

$$m = 10^{-3}$$

$$\mu = 10^{-6}$$

$$n = 10^{-9}$$

$$p = 10^{-12}$$

$$k = 10^3$$

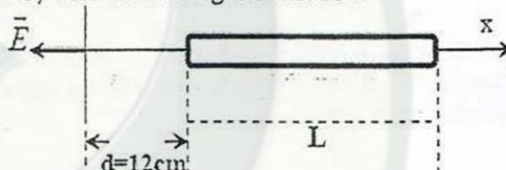
$$M = 10^6$$

$$G = 10^9$$

$$T = 10^{12}$$

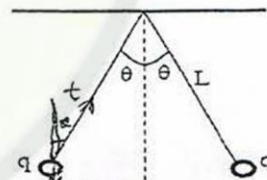
ملاحظة: يرجى نقل رمز الإجابات الصحيحة إلى الجدول المخصص للإجابات في آخر الصفحة الثانية علماً بأن عدد الأسئلة 11 سؤال.

Q.1) A rod of length L , as shown below, has a uniformly distributed charge Q . If $Q = 1.1 \times 10^{-11} \text{ C}$ and the magnitude of the electric field produced at $x = 0$ is $E = 2.58 \text{ N/C}$, what is the length of the rod?



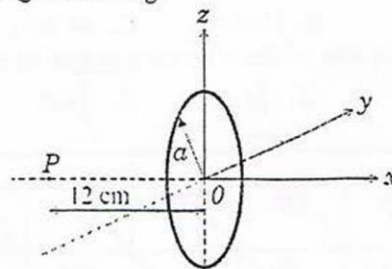
- A. 0.32 m **B. 0.2 m** C. 0.44 m D. 0.12 m

Q.2) Two identical negatively charged spheres each having a mass of $3 \times 10^{-2} \text{ kg}$ and charge q , hang in equilibrium as shown below. The length of each string is $L = 0.15 \text{ m}$ and the angle $\theta = 5^\circ$. How many extra electrons ($N = q/e$) exist on each sphere?



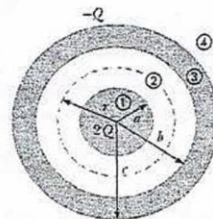
- A. 2.75×10^{11} electrons** B. 6.75×10^{16} electrons C. 3.70×10^{13} electrons D. 1.95×10^{10} electrons

Q.3) A thin ring of radius $a = 4 \text{ cm}$ contains a uniformly distributed charge Q . It is perpendicular to the x -axis with its center at the origin, as shown. The electric field at $x = -12 \text{ cm}$ on the x -axis (point P in the figure) is $E = 2.3 \times 10^3 \text{ N/C}$ directed in $+x$ -axis. Find the total charge Q on the ring?



- A. $-4.3 \times 10^{-9} \text{ C}$** B. $+3.4 \times 10^{-9} \text{ C}$ C. $-0.7 \times 10^{-9} \text{ C}$ D. $+10.7 \times 10^{-9} \text{ C}$

Q.4) A solid conducting sphere of radius a carries a net positive charge $2Q$. A conducting spherical shell of inner radius b and outer radius c is concentric with the solid sphere and carries a net charge $-Q$. find the electric field in the region 2 as shown:



- A. zero B. KQ/r^2 C. $2KQ/r^2$ D. $-KQ/r^2$

Q.5) A charge of uniform volume density (40 nC/m^3) fills a cube with 8.0 cm edges. What is the total electric flux through the surface of this cube?

- A. $2.9 \text{ N} \cdot \text{m}^2/\text{C}$ B. $2.0 \text{ N} \cdot \text{m}^2/\text{C}$ C. $2.6 \text{ N} \cdot \text{m}^2/\text{C}$ D. $2.3 \text{ N} \cdot \text{m}^2/\text{C}$

Q.6) A positively charged metal sphere A is brought into contact with an uncharged metal B. as a result:

- A. both spheres are positively charged. B. A is positively charged and B is neutral
C. A is positively charged and B is negatively charged D. A is neutral and B is negatively charged

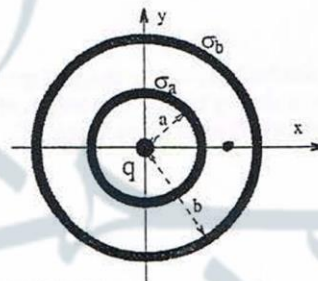
Q.7) An electric field is most directly related to:

- A. the momentum of a test charge B. the kinetic energy of a test charge
C. the potential energy of a test charge D. the force acting on a test charge

Q.8) Two thin spherical shells, one with radius R and the other with radius $2R$, surround an isolated charged point particle. The ratio of the number of field lines through the larger sphere to the number through the smaller is:

- A. 1 B. 2 C. 4 D. $\frac{1}{2}$

Q.9) A point charge $q = 6 \mu\text{C}$ is at the origin. Spherical surfaces with charge densities $\sigma_a = -0.3 \mu\text{C/m}^2$ and $\sigma_b = 0.3 \mu\text{C/m}^2$ and radii $a = 2 \text{ m}$ and $b = 4 \text{ m}$ are centered at the origin. Find the electric field at a point along the x-axis at $x = 3 \text{ m}$.



- A. $-9.1 \times 10^{+3} \text{ N/C}$ B. $5.9 \times 10^{+3} \text{ N/C}$ C. $56 \times 10^{+3} \text{ N/C}$ D. $81 \times 10^{+3} \text{ N/C}$

Q.10) A particle (mass = 5.0 g , charge = 40 mC) moves in a region of space where the electric field is uniform and is given by $E_x = 2.5 \text{ N/C}$, $E_y = E_z = 0$. If the velocity of the particle at $t = 0$ is given by $v_y = 50 \text{ m/s}$, $v_x = v_z = 0$, what is the speed of the particle at $t = 2.0 \text{ s}$?

- A. 81 m/s B. 72 m/s C. 64 m/s D. 89 m/s

Q.11) Which one of the following is not an expression for electric charge?

- A. $\int \rho dV$ B. $\int \sigma dA$ C. $\int \lambda dl$ D. $\int \mathbf{E} \cdot d\mathbf{A}$
- volume area line area

A

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
B	A	A	C	D	A	D	D	B	D	D



Name of Student: _____ Instructor: _____

Lecture Time: C-1

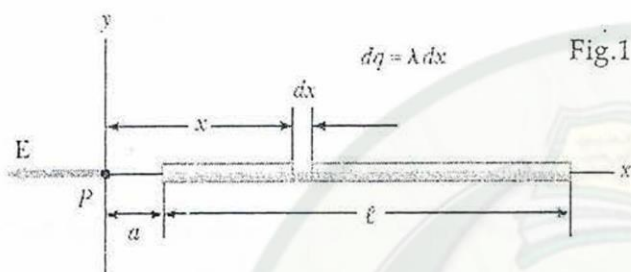
Some useful constants: $k_e = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$.

16
20

*** Important: fill in your answers in this table! ***

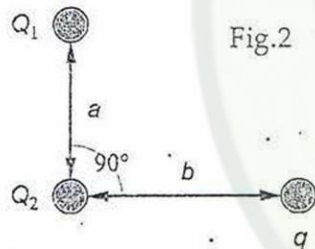
Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10
A	C	b	d	e	C	A	d	be	b

Q.1 In Figure 1, a rod of length $\ell = 30 \text{ cm}$ has a uniform positive charge per unit length $\lambda = 8 \text{ nC/m}$. Calculate the electric field at a point P that is located along the long axis of the rod and a distance $a = 5 \text{ cm}$ from one end. (Note: $1 \text{ nC} = 1 \times 10^{-9} \text{ C}$)



- a. 1234 N/C b. 8640 N/C c. 5.4 N/C d. 2.7 N/C e. zero N/C

Q.2 In Figure 2, if $a = 3.0 \text{ mm}$, $b = 4.0 \text{ mm}$, $Q_1 = 60 \text{ nC}$, $Q_2 = 24 \text{ nC}$, and $q = 80 \text{ nC}$, what is the magnitude of the total electric force on Q_2 in (milli N) ?



- a. 1080 b. 1440 c. 1800 d. 360 e. 2520

Q.3 What is the electric field in $10^6(\text{N/C})$ at the position of Q_2 in the previous Fig.2 ?

- a. 60 b. 75 c. 45 d. 15 e. 105

Q.4 A particle (mass = 4.0 g , charge = 80 mC) moves in a region of space where the electric field is uniform and is given by $E_x = -2.5 \text{ N/C}$, $E_y = E_z = 0$. If the velocity of the particle at $t = 0$ is given by $v_x = 80 \text{ m/s}$, $v_y = v_z = 0$, what is the speed of the particle at $t = 2.0 \text{ s}$?

- a. +80 m/s b. -40 m/s c. +50 m/s d. -20 m/s e. +180 m/s

Q.5 If an electric field is given by $E = 2i + 3j + 4k \text{ (N/C)}$ passing through a surface of area $A = 5 \text{ (m}^2\text{)}$ in the xz plane, the electric flux in (Nm^2/C) through this surface is

- a. 15j b. 10 c. 20 d. 10i e. 15

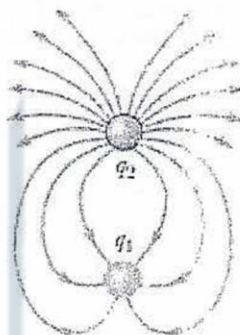
Q.6 Electric field lines in the space surrounding a charge distribution show:

- a. only the directions in which *moving* charges would accelerate when at points on those lines.
- b. only the directions in which *static* charges would accelerate when at points on those lines
- ☒ c. the directions of the electric forces that exist in space *at all times*, acting for example on a test charge
- d. that a test charge placed in this field remains *stationary* in its place
- e. that electric field lines emanate from a positive charge and end at the *same* positive charge in a *closed* path

Q.7 Figure 3 shows the electric field lines for two point charges separated by small distance then the ratio of q_1/q_2 is

- ☒ a. $-1/3$
- b. 1
- c. -3
- d. -1
- e. $-1/2$

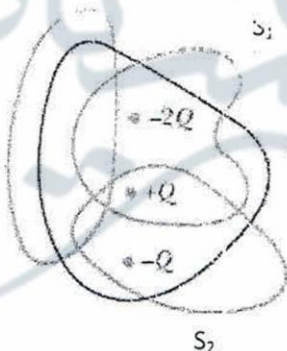
Fig.3



Q.8 Figure 4 shows 4 *closed* surfaces with charges $-2Q$, $+Q$, and $-Q$ inside. The electric flux is maximum through surface

- a. S_4
- b. S_2
- c. S_3
- ☒ d. S_1
- e. $S_1 \cup S_4$

Fig.4



Q.9 A solid plastic sphere of radius 10.0 cm has charge with uniform density throughout its volume. The electric field 5.00 cm from the center is 86.0 kN/C radially inward. Find the magnitude of the electric field 15.0 cm from the center.

- a. 38.2 kN/C
- ☒ b. 76.4 kN/C
- c. 152.8 kN/C
- d. zero kN/C
- ☒ e. 7.64 kN/C

Q.10 A long straight metal rod has a charge per unit length of 30.0 nC/m. The electric field in (N/C) at 10.0 cm from the axis of the rod is

- a. 27×10^2
- ☒ b. 54×10^2
- c. 1.0×10^{-2}
- d. 0.0×10^2
- e. 10.8×10^2



٣١.١٥.٢٢.٣٦

الاسم بالعربية: خالد عبد الجبار الجبريل موعده المحاضرة: ١١ من ٩.٥٠ في ١١ مدرس المادة: د. زيار عليان

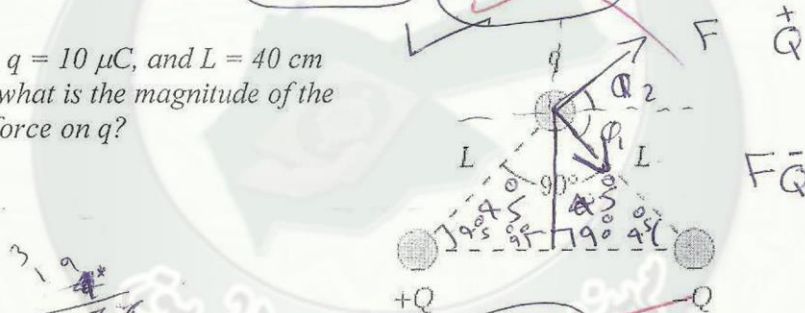
Permittivity of free space = $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$
Coulomb's Constant = $k_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$
Mass of the electron = $M_e = 9.11 \times 10^{-31} \text{ Kg}$

Mass of the Proton = $M_p = 1.67 \times 10^{-27} \text{ Kg}$
Charge of the electron = $-1.602 \times 10^{-19} \text{ C}$
Acceleration due to gravity = $g = 9.8 \text{ m/s}^2$

16

Circle the best answer of the following: (20 Mark)

- The potential (in V) in certain region is given by $V = (-24x - 30y + 16z)$ then the flux through a 2.0 m^2 portion of the yz plane is:
a) 42 b) 60 c) 34 d) 32 e) 48
- A long cylindrical shell (radius = 2.0 cm) has a charge uniformly distributed on its surface. If the magnitude of the electric field at a point 8.0 cm radially outward from the axis of the shell is 85 N/C , how much charge is distributed on a 2.0-m length of the charged cylindrical surface?
a) 0.98 nC b) 0.38 nC c) 0.19 nC d) 0.76 nC e) 0.57 nC
- If $Q = 25 \mu\text{C}$, $q = 10 \mu\text{C}$, and $L = 40 \text{ cm}$ in the figure, what is the magnitude of the electrostatic force on q ?



- a) 28 N b) 10 N c) 22 N d) 20 N e) 14 N
- A particle ($q = 4.0 \text{ mC}$, $m = 50 \text{ g}$) has a velocity of 25 m/s in the positive x direction when it first enters a region where the electric field is uniform (60 N/C in the positive y direction). What is the speed of the particle 5.0 s after it enters this region?
a) 32 m/s b) 44 m/s c) 49 m/s d) 35 m/s e) 24 m/s
- A 16 nC charge is distributed uniformly along the x axis from $x = 0$ to $x = 4 \text{ m}$. Which of the following integrals is correct for the magnitude (in N/C) of the electric field at $x = +10 \text{ m}$ on the x axis?
a) $\int_0^4 \frac{36 dx}{(10-x)^2}$ b) $\int_0^4 \frac{36 dx}{x^2}$ c) $\int_0^4 \frac{36 dx}{(x-10)^2}$ d) $\int_0^4 \frac{153 dx}{x^2}$ e) $\int_0^4 \frac{153 dx}{(10-x)^2}$
- A sphere (volume = 12 cm^3) is filled with a nonconducting material which carries a charge of 2.0 pC distributed uniformly throughout the volume. What is the magnitude of the electric field 1.0 cm from the center of the sphere?
a) 198 N/C b) 24 N/C c) 120 N/C d) 63 N/C e) 180 N/C

- [illegible]



Note: let $g=10 \text{ m/s}^2$, $\epsilon_0=8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$, $k_e=9 \times 10^9 \text{ N.m}^2/\text{C}^2$

Circle the best answer of the following:

1. (2 points) A parallel-plate capacitor has a capacitance of $30 \mu\text{F}$ and is charged with a 40 V power supply. The power supply is then removed and a dielectric of dielectric constant 2 is filled in the space between the plates. The voltage across the capacitor with dielectric is:

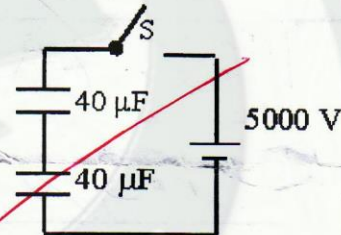
a) 20 V b) 60 V c) 120 V d) 80 V e) 5 V

2. (2 points) A charge of 12.0 nC is distributed uniformly along the x axis from $x = +2 \text{ m}$ to $x = +5 \text{ m}$. Which of the following integrals is correct for the electric potential (relative to zero at infinity) at the origin?

a) $\int_2^5 \frac{108 dx}{5-x}$ b) $\int_2^5 \frac{108 dx}{2+x}$ c) $\int_2^5 \frac{72 dx}{x}$ d) $\int_2^5 \frac{36 dx}{2-x}$ e) $\int_2^5 \frac{36 dx}{x}$

3. (2 points)

Each of the two $40 \mu\text{F}$ capacitors, as shown in the figure, is initially uncharged. How much energy is stored in the two capacitors after the switch S is closed for long time?



a) 100 J b) 300 J c) 500 J d) 250 J e) 400 J

4. (2 points) The electric potential in a certain region is described by $V(x,y,z) = 2xy - 4x^2 + 6y^2$. Find the magnitude of the net electric field at $x = -2$ and $y = 1$.

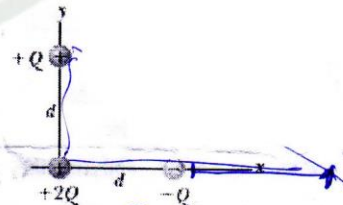
a) 14 N/C b) 9 N/C c) 11 N/C d) 13 N/C e) 16 N/C

5. (2 points) A carbon resistor has a resistance of 40Ω at a temperature of 20°C . What is its resistance at a temperature of 120°C (The temperature coefficient of resistivity for carbon is $-5.0 \times 10^{-4} / ^\circ\text{C}$)

a) 42Ω b) 32Ω c) 38Ω d) 10Ω e) 11Ω

6. (2 points)

In the figure shown, three point charges, the first $(+2Q)$ is at the origin, the second $(-Q)$ is located along the x axis at $x = d$, and the third charge $(+Q)$ located along the y axis at $y = d$. The symbolic expression for the electric potential at the point P located along the x axis at $x = +2d$.



a) $0.45 \left(\frac{kQ}{d} \right)$ b) $3.55 \left(\frac{kQ}{d} \right)$ c) $\sqrt{5} \left(\frac{kQ}{d} \right)$ d) $5 \left(\frac{kQ}{2d} \right)$ e) $2.45 \left(\frac{kQ}{d} \right)$

7. (2 points)

Electric charges flow through the wire shown in Figure, the cross-sectional areas are $A_1=6 \text{ mm}^2$ and $A_2= 3 \text{ mm}^2$ respectively. What is the drift speed of the charges in the narrow region of the wire if their speed is 0.04 m/s in the wider region?



- a) 0.02 m/s b) 0.08 m/s c) 0.32 m/s d) 0.16 m/s e) 0.04 m/s

8. (1 point) When a positive charge is released and moves along an electric field line, it moves to a position of

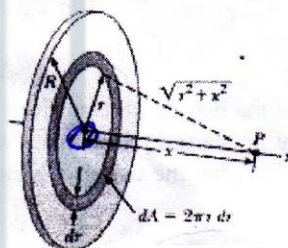
- a) lower potential and lower potential energy. b) higher potential and lower potential energy.
c) higher potential and higher potential energy. d) lower potential and higher potential energy.
e) greater magnitude of the electric field.

9. (3 points) A uniformly charged disk has radius $R=60 \text{ cm}$ and surface charge density $\sigma = 12 \text{ nC/m}^2$. Find the electric potential at the points P_1, P_2 located along the perpendicular central axis (x-axis) of the disk. $x(P_1)=0 \text{ cm}$, $x(P_2)=80 \text{ cm}$,

$V(P_1) = \dots 407 \dots \text{V}$

$V(P_2) = \dots 135.7 \dots \text{V}$

$|\Delta V| = \dots 271.3 \dots \text{V}$



10. (2 points) A 8.4 V potential difference is maintained across a 3.0 m length wire that has a diameter of 0.80 mm . If the wire is made of material that has a resistivity of $7.0 \times 10^{-8} \Omega \cdot \text{m}$.

- (A) What is the resistance of the wire. (B) What is the current in the wire?

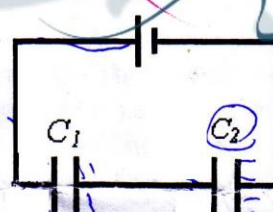
(A) The resistance =

(B) The current =

11. (1 point)

Two parallel-plate capacitors are connected in series to a battery as shown in the figure. If the separation distance in capacitor C_2 is decreased then :

- a) Charge on C_1 increases.
b) Charge on C_1 decreases
c) Charge on C_1 remains the same.
d) Charge on C_1 increases or decreases depending on the voltage of the battery.
e) Charge on C_1 increases or decreases depending on the dielectric constant of the dielectric.





الاسم بالعربية: موعده المحاضرة (١١ - ١٢) مدرس المادة: د. أيمن مشدوق

Permittivity of free space = $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$.
Coulomb's Constant = $k_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$.
Mass of the electron = $M_e = 9.11 \times 10^{-31} \text{ Kg}$.

Mass of the Proton = $M_p = 1.67 \times 10^{-27} \text{ Kg}$.
Charge of the electron = $-1.6 \times 10^{-19} \text{ C}$.
Acceleration due to gravity = $g = 9.8 \text{ m/s}^2$

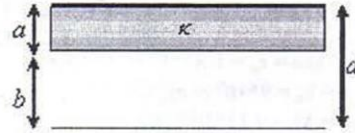
Circle the best answer of the following: (20 Mark) + (1 questions bonus).

- A current of 0.3 A is passed through a lamp for 2 minute using a 6 V power supply. The energy dissipated by this lamp during this time (in J) is:
a) 216 **b) 20** c) 3.6 d) 126 e) 108
- If a 4 μF capacitor charged to 40 V and a 6 μF capacitor charged to 15 V are connected to each other, with the positive plate of each connected to the negative plate of the other, then the final charge on the 4 μF capacitor is equal to:
a) 10 μC **b) 60 μC** c) 70 μC d) 28 μC e) 42 μC
- How many electrons pass through a 20 Ω resistor in 5 min if there is a potential drop of 30 volts across it?
a) 9.4×10^{21} b) 7.5×10^{21} **c) 2.8×10^{21}** d) 1.1×10^{21} e) 5.6×10^{21}
- A proton (mass = $1.67 \times 10^{-27} \text{ kg}$, charge = $1.60 \times 10^{-19} \text{ C}$) moves from point A to point B under the influence of an electrostatic force only. At point A the proton moves with a speed of 60 km/s. At point B the speed of the proton is 80 km/s. Then the potential difference $V_B - V_A$ is equal to:
a) +15 V **b) -15 V** c) +33 V d) -33 V e) 20 V
- Over a certain region of space, the electric potential is given by $V(x,y) = (-5xy^2 + 3xy - 6) \text{ volt}$, then the electric field vector at the point (-2,1) is equal to:
a) $2\hat{i} - 14\hat{j}$ b) $8\hat{i} + 2\hat{j}$ c) $8\hat{i} - 26\hat{j}$ d) $20\hat{i} + 14\hat{j}$ e) $20\hat{i} + 23\hat{j}$
- A charge $Q = 0.18 \mu\text{C}$ is uniformly distributed along the x-axis from $x = -4 \text{ m}$ to $x = +2 \text{ m}$, then the electric potential (relative to zero at infinity) at the point $x = 10 \text{ m}$, on the x-axis is equal to:
a) 906 V **b) 60 V** c) 49 V d) 82 V e) 151 V
- A charge of 20 nC is distributed uniformly along the y axis from $y = 0$ to $y = 4 \text{ m}$. Which of the following integrals is correct for the electric potential (relative to zero at infinity) at the point $x = +3 \text{ m}$ on the x axis?

a) $\int_0^4 \frac{45 dx}{(y^2 + 9)^{1/2}}$ b) $\int_0^4 \frac{180 dx}{(y^2 + 9)}$ c) $\int_0^4 \frac{180 dx}{(y^2 + 9)^{1/2}}$ d) $\int_0^4 \frac{45 dx}{(y^2 + 9)}$

8.

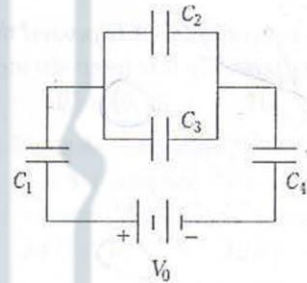
A parallel-plate capacitor with a plate separation d has a capacitance $C_0 = 5 \mu\text{F}$ in the absence of a dielectric. When a slab (شريحة) of dielectric material of dielectric constant $\kappa = 3$ and thickness $a = d/3$ is inserted between the plates as shown in the figure, the capacitance becomes:



- a) $15 \mu\text{F}$ b) $3.9 \mu\text{F}$ c) $6.4 \mu\text{F}$ d) $8.3 \mu\text{F}$ e) $3 \mu\text{F}$

9.

In the figure shown, if $C_1 = 20 \mu\text{F}$, $C_2 = 10 \mu\text{F}$, $C_3 = 14 \mu\text{F}$, $C_4 = 30 \mu\text{F}$, and $V_0 = 45 \text{ V}$, then the energy stored by C_4 is equal to:



- a) 3.8 mJ b) 2.7 mJ c) 3.16 mJ d) 2.16 mJ e) 8.1 mJ

10. A nonconducting sphere of radius 10 cm is charged uniformly with a density of $0.30 \mu\text{C}/\text{m}^3$. Then the magnitude of the potential difference between the center and a point 4.0 cm away is equal to:

- a) 12 V b) 6.8 V c) 3.0 V d) 4.7 V e) 2.2 V

11. The quantity of charge q (in C) that passes through a surface varies with time according to the equation $q = 6t^3 + 3t^2 - 1$, (t in s). The instantaneous current (in A) through the surface at $t = 0.5 \text{ s}$

- a) 49.5 b) 127.5 c) 10 d) 7.5 e) 3



مدرس المادة:

موعد المحاضرة:

الاسم بالعربية:

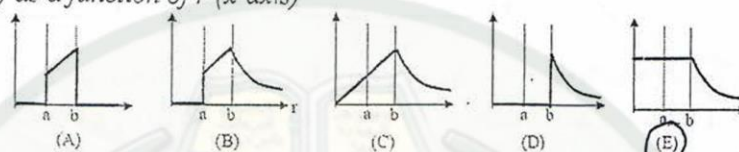
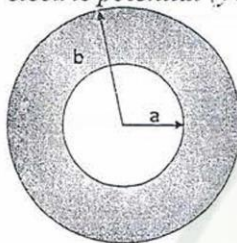
Permittivity of free space $= \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$

Coulomb's Constant $= k_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$

Circle the best answer of the following: (20 Mark).

1.

A conducting spherical shell of inner radius a and outer radius b carries a net charge $+Q$. Which one of the following graphs given below is correctly describes the magnitude of electric potential (y -axis) as a function of r (x -axis)



a) A

b) B

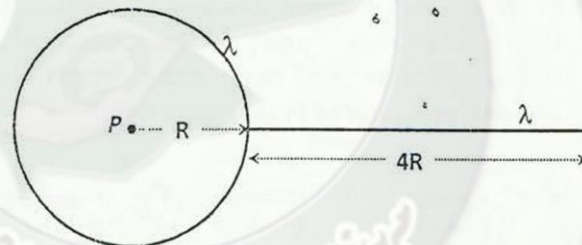
c) C

d) D

e) E

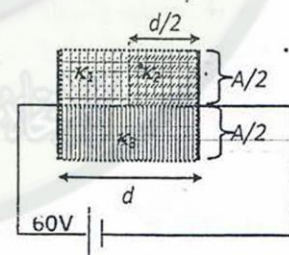
2.

A ring of radius $R=10 \text{ cm}$ and a rod of length $l=4R$, are arranged as shown in figure below, find the magnitude of the electric potential at the point P, if both they (the ring and the rod) have the same uniform linear charge density $\lambda=8 \text{ nC/m}$



3.

A parallel-plate capacitor of plate area $A=10 \text{ cm}^2$ and plate separation $d=2 \text{ mm}$ is constructed using three insulators of $\kappa_1=4.0$, $\kappa_2=6.0$ and $\kappa_3=8.2$, as shown in the figure. Find the voltage across the dielectric of constant κ_2



a) 42 V

b) 18.0 V

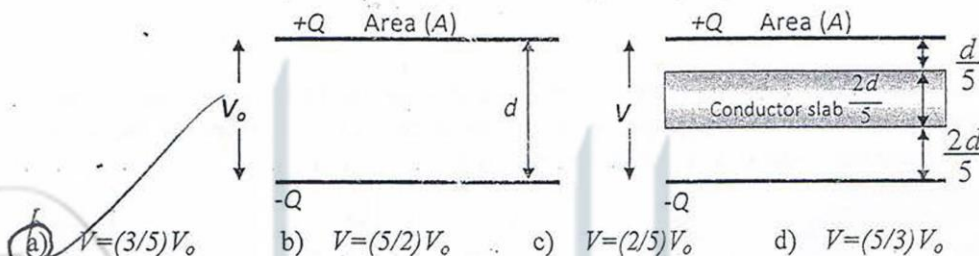
c) 24.0 V

d) 21.0

e) 4.5 V

E 1

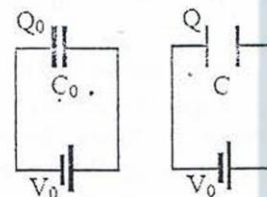
4. A metal wire has a resistance of 7.0Ω at a temperature of 20.0°C . If the same wire has a resistance of 7.8Ω at 100.0°C , what is the resistance of this wire when its temperature is -20°C ?
- a) 0.70Ω b) 9.66Ω c) 7.8Ω **d) 6.5Ω**
5. Two spherical conductors of radii $R_1 = 4 \text{ cm}$ and $R_2 = 12 \text{ cm}$ are separated by a very large distance and each of them carry the same positive charge $Q_1 = Q_2 = 10 \mu\text{C}$. If the spheres are connected by a conducting wire, Find the surface charge density (in $\mu\text{C}/\text{m}^2$) of the larger sphere.
- a) 248.7** b) 27.6 c) 127.3 d) 746 e) 82.9
6. What is the relationship between V_0 and V in the system (of capacitors) shown below?



7. A capacitor has vacuum in the space between its plates. If you increase the plates potential difference by a factor of 2, what happens to the capacitance?

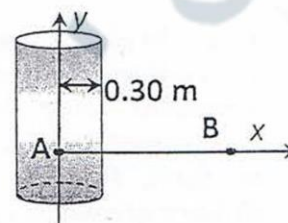
- a) It increases by a factor of two** c) It remains the same
b) It decreases by a factor of two d) we can't know

8. A parallel-plate capacitor C is fully charged by a battery V . Now, the distance between the parallel plates is decreased while the battery remains connected to the same battery. What happens to the capacitance C_0 , charge Q_0 , and energy U_0 stored in the capacitor?



- a) C increases, Q increases, U increases. b) C decreases, Q stay the same, U stay the same **c) C decreases, Q decreases, U decreases** d) C decreases, Q increases, U decreases e) C decreases, Q stay the same, U decreases

9. The axis of a very long conducting cylinder (of radius $R = 0.3 \text{ m}$) is on the y -axis (see figure). The cylinder has surface charge density $\sigma = 4.0 \mu\text{C}/\text{m}^2$. The magnitude of the potential difference (in kV) between the point A (at $x = 0 \text{ m}$) and B (at $x = 3 \text{ m}$) shown in the figure



- a) 351.2 b) 39 **c) 162.7** d) 148.5 e) 312.2
10. Nicrom wire of radius 0.32 mm , length 0.5 m , and resistivity $150 \mu\Omega/\text{cm}$. If the electric power dissipated in the wire is 75.8 watt , find the electric field (in V/m) inside the wire.
- a) 18.5 b) 21.1 c) 23.7 **d) 26.6**

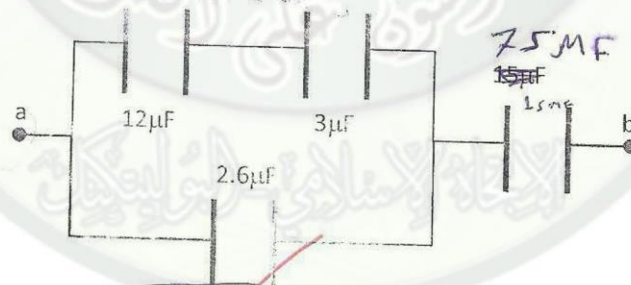


Permittivity of free space = $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$.

Coulomb's Constant = $k_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$.

Circle the best answer of the following: (20 Mark).

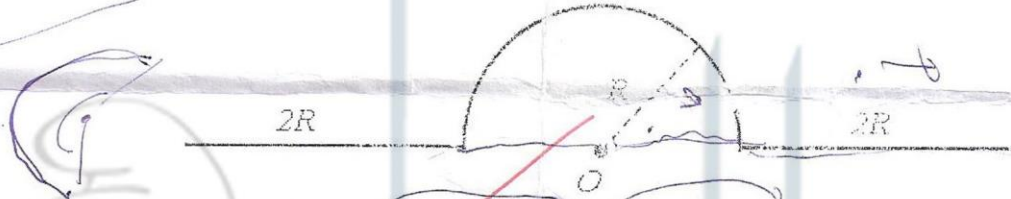
- The density of free electrons in gold is $5.90 \times 10^{28} \text{ electron/m}^3$. The resistivity of gold is $2.44 \times 10^{-8} \Omega \cdot \text{m}$ at temperature of 20°C and the temperature coefficient of resistivity is $0.004^\circ \text{C}^{-1}$. A gold wire, 1.1 mm in diameter and 29 cm long, carries a current of 650 mA. The drift speed (in m/s) of the electrons in the wire is equal to:
a) 9.4×10^{-5} b) 1.0×10^{-4} c) 7.3×10^{-5} d) 8.3×10^{-5} e) 1.2×10^{-4}
- When a parallel plate capacitor is charged to a 50 V battery its surface charge density is equal to 10 nC/m^2 , and the separation distance (in mm) between its plates is:
a) 50.0 b) 0.4 c) 4.4 d) 10.1 e) 44.3
- A 5 cm radius insulating solid sphere has a uniform volume charge density $3 \times 10^{-5} \text{ C/m}^3$ and total charge Q . The point A is located at the center and the point B at the surface of the sphere, then the magnitude of the potential difference between A and B is: [Assume $V = 0$ at infinity]
a) 1414 V b) 56540 V c) 28.7 V d) 4241 V e) 0 V
- The quantity of charge q (in coulombs) that passes through a surface of area 25 mm^2 is given by $q(t) = 4t^2 + 9t + 6$, where t in seconds. Then the current density at $t = 1 \text{ s}$ is equal to:
a) 1.6×10^6 b) 1.0×10^6 c) 40×10^3 d) 25×10^3 e) 1.0×10^6
- Capacitors A and B have the same capacitance. Capacitor A is charged so that it stores 4 J of energy and capacitor B is uncharged. The capacitors are then connected in parallel. The total stored energy in the capacitors is now:
a) 14 J b) 1 J c) 2 J d) 8 J e) 2 J
- In the figure shown below, $V_a = 20 \text{ V}$. What will be the final potential difference ΔV_{ab} if the $15 \mu\text{F}$ capacitor is filled with a dielectric of $\epsilon_r = 5$?



- a) 24.0 V b) 4.5 V c) 16.0 V d) 35.0 V e) 37.5
- An air-filled parallel-plate capacitor has a capacitance of 1 pF . The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance becomes 2 pF . The dielectric constant of the wax is:
a) 8 b) 0.5 c) 0.4 d) 4 e) 2

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8. A solid conducting sphere (radius = 5 cm) has a charge of 0.25 nC distributed uniformly on its surface. If point A is located at the center of the sphere and point B is 15 cm from the center, what is the magnitude of the electric potential difference between these two points?
a) 23 V b) 60 V c) 45 V d) 15 V **e) 30 V**
9. A wire having a uniform linear charge density 6 nC/m is bent into the shape of the figure shown below ($R=10$ cm), then the magnitude of the electric potential at point O.



- a) 229.0 V b) 237.5 V **c) 149.5 V** **d) 283.3 V** e) 214.8 V
10. A conductor of uniform radius 1.2 cm carries a current of 3 A produced by an electric field of 120 V/m, then the resistivity of the material (in $\Omega \cdot m$) is equal to:
a) 0.08 b) 55.3 c) 60.0 **d) 0.0151** e) 50.0

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